

Federal State Autonomous Educational Institution of Higher Education "Moscow Institute of Physics and Technology (National Research University)"

Bachelor's Thesis
Bachelor degree: Computer Science

« Classification of EEG data with Federated Learning »

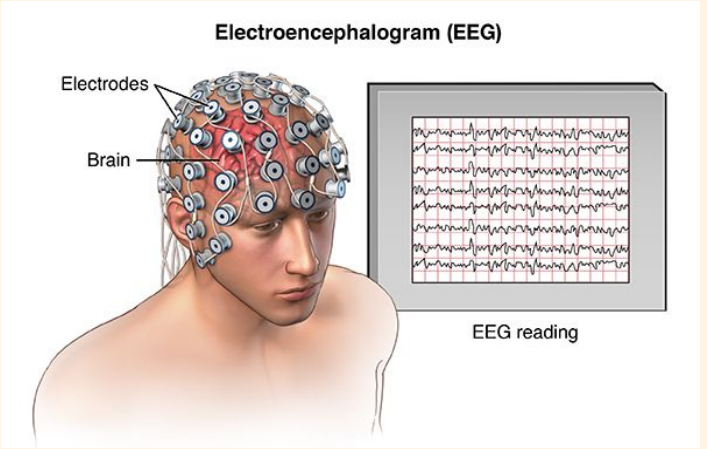
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Introduction: Basic Definitions

- **Classification** is a supervised machine learning method where the model tries to predict the correct label of a given input data.
- **Federated Learning** is a decentralized machine learning technique that doesn't require the data to be transferred to a central unit but rather to be kept on the users' devices.
- **Electroencephalogram (EEG)** is a test that records the brain activity using electrodes that are sticked on the scalp of the head, the electrodes can be described as metal discs with thin wires.



Y. Shen. 2020

Introduction: Classification Metrics

- Accuracy = (TP + TN) / (TP + FP + FN + TN)
- F1 = 2 · TP / (2 · TP + FP + FN)
- Kappa = $\frac{2 \cdot (TP \cdot TN - FP \cdot FN)}{(TP + FP) \cdot (FP + TN) + (TP + FN) \cdot (FN + TN)}$
- AUROC (Area Under Receiver Operating Characteristic Curve): 1/ ROC Curve Area*

	P' (Predicted)	N' (Predicated)
P (Actual)	True Positive	False Negative
N (Actual)	False Positive	True Negative

*ROC graph has the x-axes as the False positive rate and the y-axes as the True Positive Rate

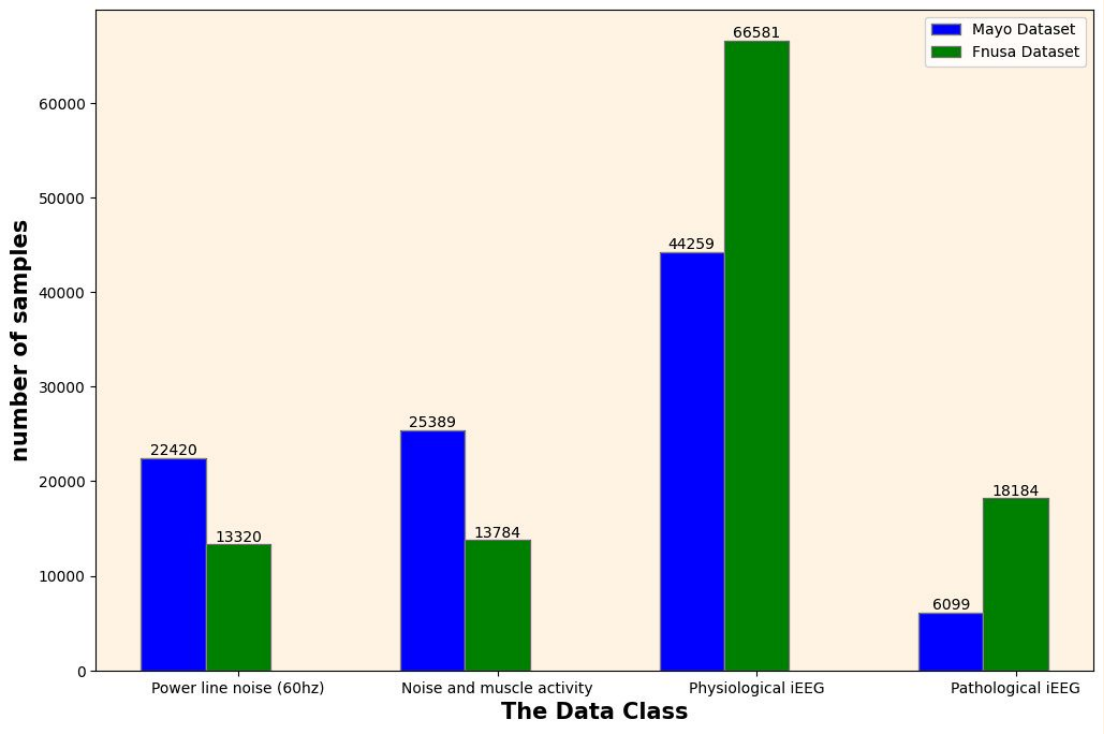
Introduction: The problem

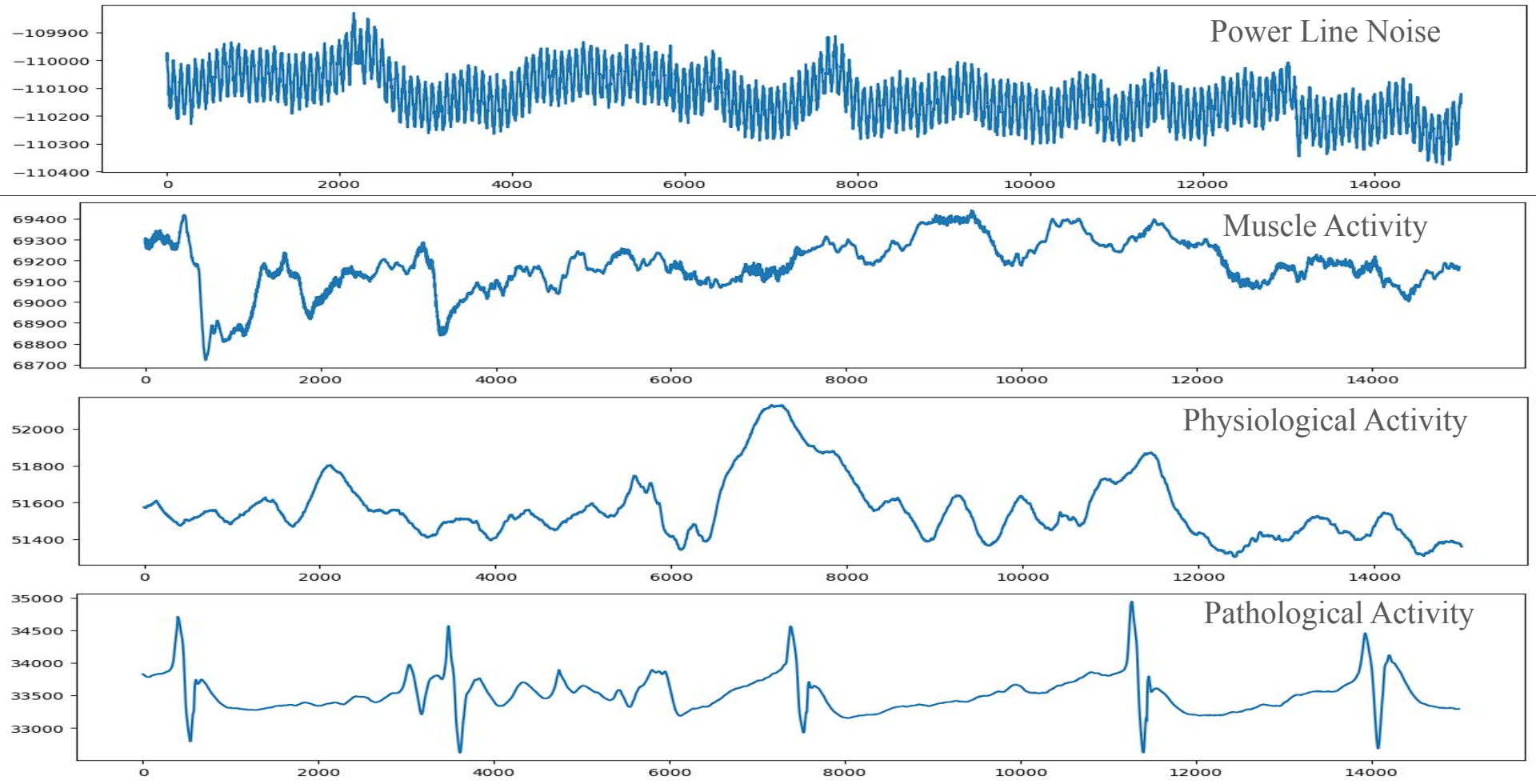
1. Using the data from different clients to train the classification model without exposing it to third parties or storing in a centralized server.

2. Mitigating the problems that appear with non-heterogeneous data.

Methods: The Dataset

- The dataset was collected from two Institutions, St Anne's University Hospital (FNUSA) and Mayo Clinic (Mayo).
- St. Anne's University Hospital: 11 patients
- Mayo Clinic: 25 patients
- DRE (Drug Resistant Epilepsy)




Power Line Noise
−109900
−110000
−110100
−110200
−110300
−110400
0
2000
4000
6000
8000
10000
12000
14000
Muscle Activity
69400
69300
69200
69100
69000
68900
68800
68700
Physiological Activity
52000
51800
51600
51400
Pathological Activity
35000
34500
34000
33500
33000

Methods: Data preprocessing and training scheme

- Single Channel input with segments of 15000 samples length.
- Each iEEG segment was converted into a spectrogram and further normalized by calculating z-scores for each frequency band.
- Only the first 200 frequency bins are kept for training. To reduce the dimensionality of the data.
- The data was undersampled to 13784 samples, in order to avoid biases.
- Out-of-institution scheme was used in the testing process.
- Different acquisition systems and different behavioral state (awake and asleep)

Methods: Data preprocessing and training scheme

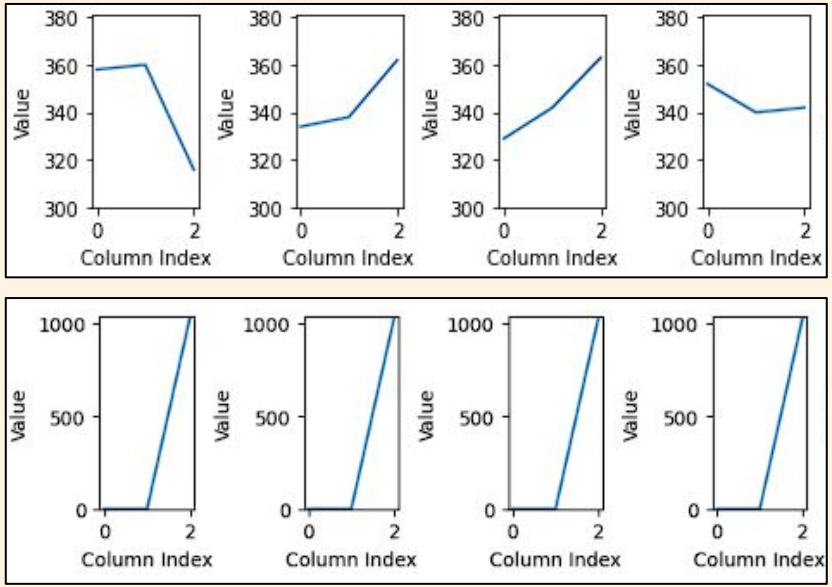
- Three scenarios were tested:
  1- IID

  2- Label Distribution Skew (Weak Non IID)

  3- Quantity-based Label Imbalance (Extreme Non IID)

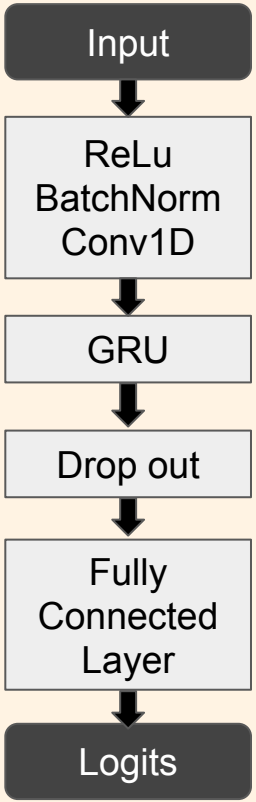
- Two optimizers were used:
  1- Adam

  2- Scaffold Optimizer.



Methods: The Model Architecture

- Conv1D with 200 * 3 kernel size
- Batch normalization and rectified linear units were attached.
- The resulting vectors were fed into the GRU.
- Drop out with probability 0.5.
- Fully connected linear layer with 3 as the number of outputs.



Methods: Federated Learning Algorithms

- Scaffold and FedAvg are the two main algorithms used in this thesis.

- Scaffold has the following three main ideas:

1. Local updates to the client model.

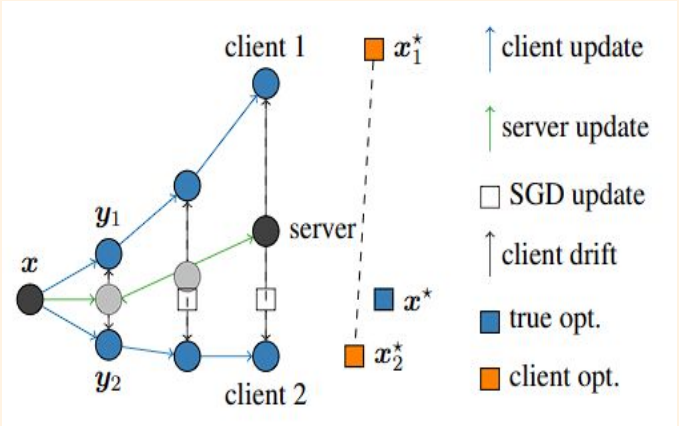
$$y_i \leftarrow y_i - \eta_l \left(g_i(y_i) + c - c_i \right)$$

2. Local updates to the client control variate

$$c_i^+ = g_i(x) \; || \; c_i - c + \frac{1}{k\eta_l}(x - y_i)$$

3. Aggregating the updates

$$x \leftarrow x + \frac{\eta_g}{|S|} \sum_{i \in S} (y_i - x) \qquad c \leftarrow c + \frac{1}{N} \sum_{i \in S} (c_i^+ - c_i)$$



S. P. Karimireddy 2021

N: total num., S: sampled num, x: agg. server model, c_i, c: local and global control variates

Results

			PREDICTED					
EXPECTED	FedAvg		Noise and muscle activity		Physiological iEEG		Pathological iEEG	
	Noise and muscle activity	IID	23922	22437	1090	2261	375	690
		Weak Non-IID	22697		2492		198	
		Extreme Non-IID	***4***		24779		606	
	Physiological iEEG	IID	7218	3341	31420	35389	5620	5527
		Weak Non-IID	5283		34482		4493	
		Extreme Non-IID	1		***43450***		805	
	Pathological iEEG	IID	65	58	390	618	5644	5423
		Weak Non-IID	55		745		5299	
		Extreme Non-IID	0		2785		***3314***	

Results

	FedAvg			Scaffold			
	ADAM			ADAM			Scaffold optimizer
Data Setting	IID	Weak Non IID	Extreme Non IID	IID	Weak Non IID	Extreme Non IID	Extreme Non IID
Learning Rate	0.001	0.001	0.001	0.001	0.001	0.0003	0.0001
Number of Rounds	15	15	100	25	25	100	100
F1 Score	0.76	**0.78**	0.45	0.66	0.7	0.31	**0.5**
Kappa Score	0.67	**0.7**	0.14	0.53	0.57	0.05	**0.3**
AUROC	0.94	**0.94**	0.57	0.94	0.93	0.7	**0.69**

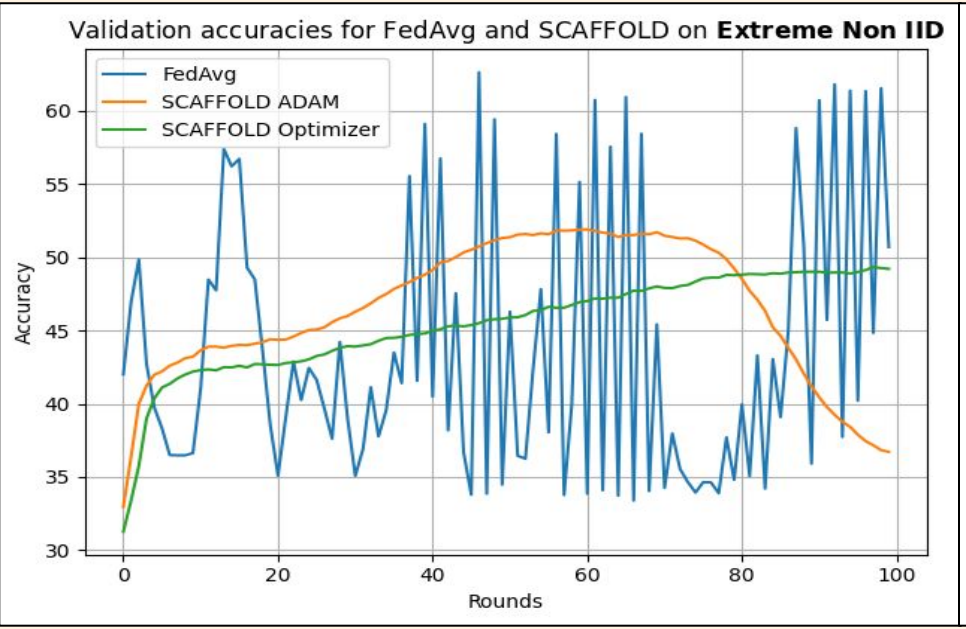
The Kappa Score achieved by the baseline is 0.71 with 0.93 AUROC and 0.77 F1 score

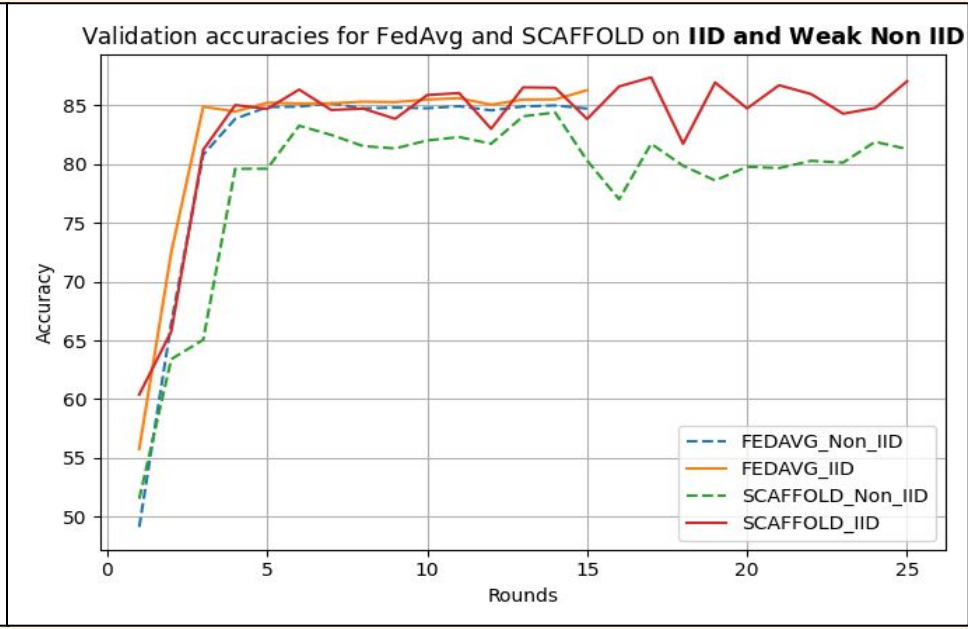
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Results


Validation accuracies for FedAvg and SCAFFOLD on **Extreme Non IID**
FedAvg
SCAFFOLD ADAM
SCAFFOLD Optimizer
Accuracy
Rounds
30
35
40
45
50
55
60
0
20
40
60
80
100


Validation accuracies for FedAvg and SCAFFOLD on **IID and Weak Non IID**
FEDAVG_Non_IID
FEDAVG_IID
SCAFFOLD_Non_IID
SCAFFOLD_IID
Accuracy
Rounds
50
55
60
65
70
75
80
85
0
5
10
15
20
25

Conclusions

- FedAvg algorithm achieves results that are on the same level of performance as the centralized model.
- FedAvg is a better choice than SCAFFOLD when the clients have very similar data distributions or come from the same distribution.
- SCAFFOLD algorithm mitigates the performance issues accompanied with non-heterogeneous data but doesn’t solve the problem.
- The significance of this study is that we used a separate dataset for testing the performance of the models.